Candidate Seat No: _____

Charotar University of Science and Technology

Devang Patel Institute of Advance Technology and Research

Department of Computer Science and Engineering

Department of Computer Engineering

Department of Information Technology

Subject: CSUE201 – Foundation of Data Science and Analytics

CIE – 2: Test of UNIT 2

Semester: 4 | Maximum Marks: 20

Date: 26/02/2026 | Time: 09:10 AM to 10:10 AM

Instructions:

i. Attempt all the questions.
ii. Figures to the right indicates full marks.
iii. Give appropriate example whenever if required.
iv. Use the provided dataset only.

Answer the following questions				
C02	A	1.	A real estate analytics startup is preparing to build a predictive model for estimating house prices. Before constructing the model, the data scientist performs an exploratory statistical study to understand how different variables influence price. The company suspects that either: • The size of the house (in square feet), or • The distance of the house from the city center has a stronger linear relationship with the selling price. The following data of 5 recently sold houses is collected:	**[05]**

House	**Area (sq.ft)**	**Distance (km)**	**Price (Lakhs)**
1	1000	15	50
2	1500	12	60
3	2000	10	75
4	2500	8	90
5	3000	5	110

Answer the following:

(a) Identify and apply an appropriate statistical measure to determine which independent variable has a stronger linear association with house price. Show all necessary calculations.

(b) Based on your computed results, recommend which variable should be retained for predictive modeling. Provide mathematical justification.

(c) Explain one limitation of relying only on this statistical measure for feature selection in real-world datasets.

C02 | E | 2. | **[05]**

A telecom company is developing a **machine learning model to predict customer churn**. During **Exploratory Data Analysis (EDA)**, the data scientist applies the **Chi-Square Test (Filter Method)** to determine whether the feature **"Internet Plan Type"** is significantly associated with customer churn.

From a sample of 120 customers, the following data is collected:

Internet Plan	Churn (Yes)	Churn (No)	Total
Basic	30	20	50
Standard	25	25	50
Premium	5	15	20
Total	60	60	120

The test is to be conducted at:

$$\alpha = 0.05$$

Degree of Freedom (df)	χ^2 Critical Value
1	3.84
2	5.99
3	7.81
4	9.49

Tasks:

(a) Calculate the expected frequencies for each cell.

(b) Compute the Chi-Square statistic using:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

(c) Determine the degree of freedom.

(d) Compare the calculated χ^2 value with the critical value and conclude whether "Internet Plan Type" should be selected as an important feature using the filter method.

C02 | E | 3. | **[05]**

A digital payment company is analyzing **daily transaction amounts (in ₹ thousands)** to detect suspicious activities during **Exploratory Data Analysis (EDA)**.

			The transaction amounts recorded for 12 customers in a day are: $5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 18, 85$ The data scientist decides to detect outliers using the **Interquartile Range (IQR) method**. **Tasks:** **(a)** Compute Q1, Q3 and IQR. **(b)** Determine the lower and upper bounds using: $\text{Lower Bound} = Q1 - 1.5(IQR)$ $\text{Upper Bound} = Q3 + 1.5(IQR)$ **(c)** Identify the outlier(s), if any. **(d)** The transaction value ₹85 belongs to a verified corporate client who regularly performs bulk payments. Should it automatically be treated as fraud? Justify your answer in the context of outlier's vs anomalies.	
C02	C	4.	A national education analytics company is conducting a large-scale study to analyze the **digital learning behavior of students across India**. The dataset contains information about 50,000 students from different: • States • Urban and rural regions • Government and private schools • Different grade levels Due to cost and time constraints, the company cannot collect data from all students and must use sampling techniques. Answer the following: **(a)**If every student in the population has an equal chance of being selected using a lottery method, which sampling technique is used? Explain briefly. **(b)**If the company selects every 100th student from an alphabetically ordered list, which sampling method is being applied? Mention one limitation of this method. **(c)**If students are first divided into **Urban and Rural groups**, and then random samples are taken from each group proportionally, which sampling technique is this? Why is it preferred in this case? **(d)**If entire schools are randomly selected and all students from those selected schools are surveyed, which sampling method is this? **(e)**Differentiate between Stratified Sampling and Cluster Sampling in one clear point	**[05]**